Space Observation Intelligence Platform

Explore, Plan & Learn

"Discover what you can observe, plan the best time to observe it, and understand why celestial events occur."

1. Project Scope

The Space Observation Intelligence Platform is a data-driven web application that enables users to explore the night sky based on their location, date, and time. It integrates astronomical data from trusted sources, analyzes celestial information, and presents it through interactive visualizations and educational explanations.

Objectives:

- **Explore:** Identify visible celestial objects (planets, Moon, stars, constellations, events).
- **Plan:** Recommend suitable observation times and visibility info.
- **Learn:** Explain celestial phenomena with interactive visuals and simplified content.

2. Problem Statement

Astronomical information is scattered, technical, or focused only on visualization. Beginners struggle with questions like:

- What can I observe tonight?
- When is the best time?
- Why does the Moon change its phase?
- Why are planets visible in different positions?

This project integrates these aspects into one user-friendly platform.

3. Objectives

- Collect astronomical data from reliable sources.
- Organize and store data efficiently.
- Provide interactive exploration.
- Recommend observation windows.
- Present educational explanations.
- Build an intuitive dashboard.

4. Target Users

- Students
- Teachers
- Amateur astronomers
- Astronomy clubs
- Science enthusiasts
- Beginners

5. Functional Requirements

Explore Module

- Search by location, date, time
- Display planets, Moon phase, stars, constellations, events

Plan Module

- Recommend best observation time
- Show rise/set times, directions, observation info

Learn Module

- Explain Moon phases, eclipses, planetary visibility/colors
- Interactive animations, basic facts

Analytics Module

- Planet visibility trends
- Moon phase analysis
- Event statistics
- Interactive charts

6. Non-Functional Requirements

- User-friendly interface
- Fast response time
- Responsive design
- Reliable astronomical calculations

- Easy to extend

7. Module Specifications

Module 1 – Data Collection

- Retrieve, process, store astronomical data
- Input: APIs, astronomy libraries
- Output: Structured database

Module 2 – Explore

- Generate current night sky info
- Output: Visible planets, Moon info, events

Module 3 – Plan

- Observation recommendations
- Output: Best viewing time, visibility details, schedule

Module 4 – Learn

- Educational explanations, interactive visuals
- Output: Animations, simplified explanations, object info

Module 5 – Analytics

- Analyze datasets
- Output: Graphs, trends, dashboards

8. Technology Stack

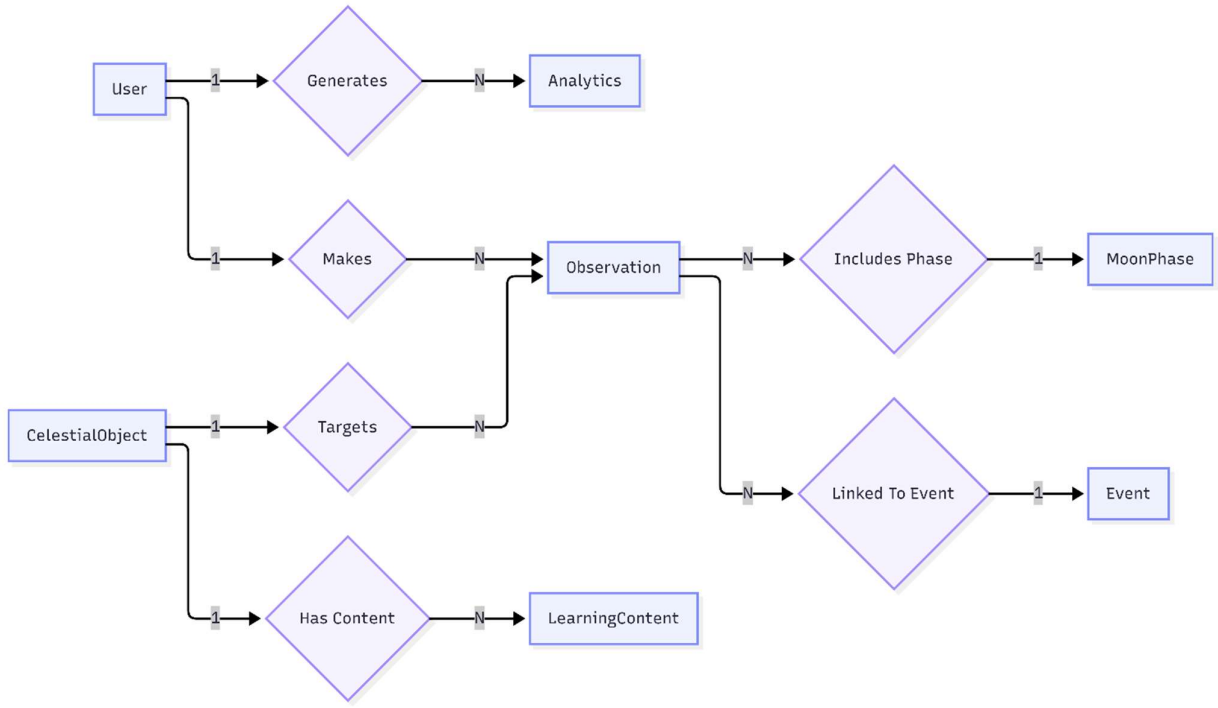
- **Programming:** Python
- **Database:** MySQL
- **Data Processing:** Pandas, NumPy
- **Astronomy:** Skyfield, Astropy
- **Frontend:** Streamlit
- **Visualization:** Plotly, Matplotlib
- **Tools:** VS Code, Git, GitHub

9. Future Scope

- AI-powered astronomy assistant

- Weather integration
- Light pollution analysis
- Astrophotography recommendations
- Deep-sky object explorer
- Space mission analytics
- Mobile app

ERD:



Use Case:

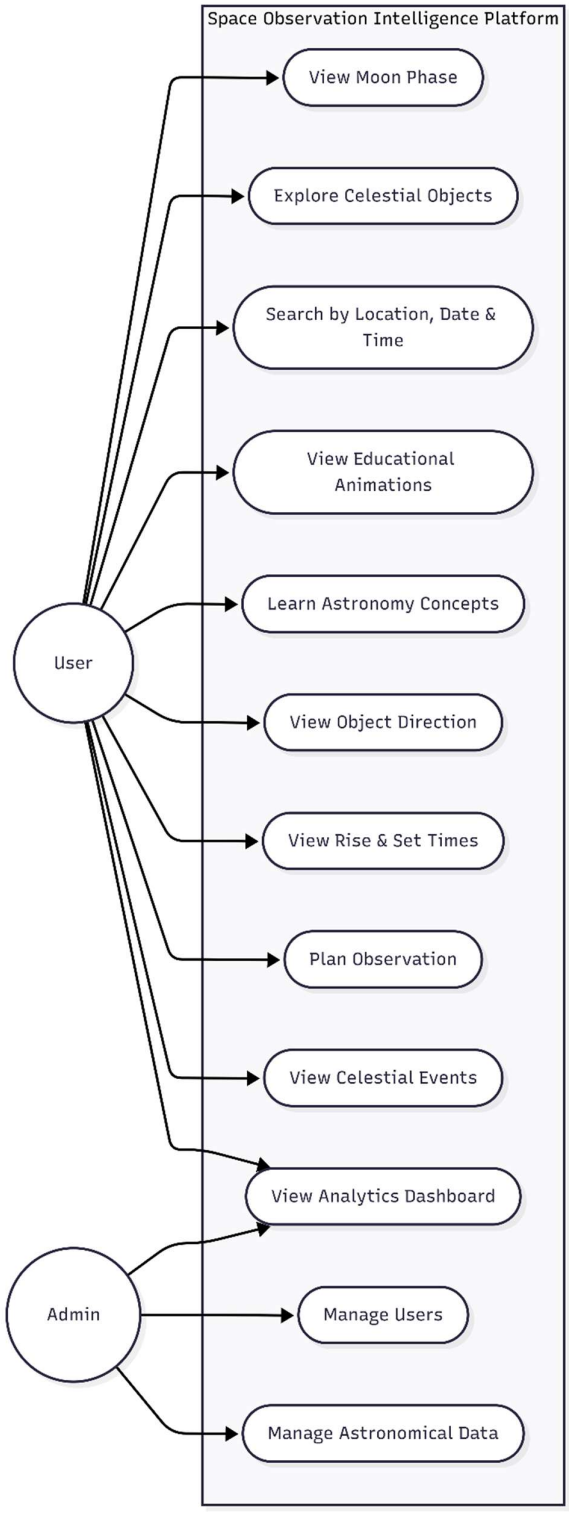
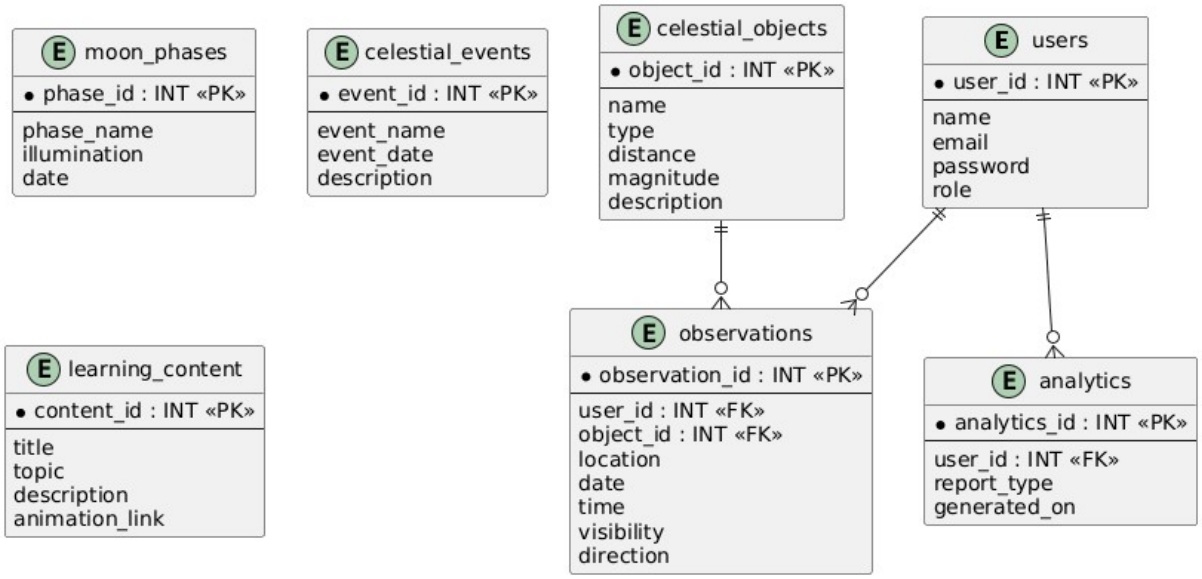

Space Observation Intelligence Platform
View Moon Phase
Explore Celestial Objects
Search by Location, Date & Time
View Educational Animations
Learn Astronomy Concepts
User
View Object Direction
View Rise & Set Times
Plan Observation
View Celestial Events
View Analytics Dashboard
Admin
Manage Users
Manage Astronomical Data

Table Structure:



Database Schema

Entity / Table Name	Primary Key	Foreign Key(s)	Other Attributes / Columns
moon_phases	phase_id (INT)	*None*	phase_name, illumination, date
celestial_events	event_id (INT)	*None*	event_name, event_date, description
learning_content	content_id (INT)	*None*	title, topic, description, animation_link
celestial_objects	object_id (INT)	*None*	name, type, distance, magnitude, description

Entity / Table Name	Primary Key	Foreign Key(s)	Other Attributes / Columns
users	user_id (INT)	*None*	name, email, password, role
observations	observation_id (INT)	user_id (INT), object_id (INT)	location, date, time, visibility, direction
analytics	analytics_id (INT)	user_id (INT)	report_type, generated_on